



AHoJ-DB: A PDB-wide Assignment of apo & holo Relationships Based on Individual Protein–Ligand Interactions

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Abstract

A single protein structure is rarely sufficient to capture the conformational variability of a protein. Both bound and unbound (holo and apo) forms of a protein are essential for understanding its geometry and making meaningful comparisons. Nevertheless, docking or drug design studies often still consider only single protein structures in their holo form, which are for the most part rigid. With the recent explosion in the field of structural biology, large, curated datasets are urgently needed. Here, we use a previously developed application (AHoJ) to perform a comprehensive search for apo-holo pairs for 468,293 biologically relevant protein–ligand interactions across 27,983 proteins. In each search, the binding pocket is captured and mapped across existing structures within the same UniProt, and the mapped pockets are annotated as apo or holo, based on the presence or absence of ligands. We assemble the results into a database, AHoJ-DB (www.apoholo.cz/db), that captures the variability of proteins with identical sequences, thereby exposing the agents responsible for the observed differences in geometry. We report several metrics for each annotated pocket, and we also include binding pockets that form at the interface of multiple chains. Analysis of the database shows that about 24% of the binding sites occur at the interface of two or more chains and that less than 50% of the total binding sites processed have an apo form in the PDB. These results can be used to train and evaluate predictors, discover potentially druggable proteins, and reveal protein- and ligand-specific relationships that were previously obscured by intermittent or partial data. Availability: www.apoholo.cz/db

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Introduction

Recent advances in methodology and processing power have slingshot structural biology to the forefront of biological research, bridging the gap between 1D and 3D data.^{1,2} Models of millions of protein structures are now available^{3,4} while the most widely used method in computational biology, similarity search, has been reinvented to enable ultra-fast searches in the new and vast 3D structure space.⁵ In line with the paradigm that structure is

better conserved than sequence,^{6,7} the newly acquired ability to perform fast and accurate structure searches, “already outperforms sequence-based alignment methods in many ways”, even when using moderately accurate models.^{8,9}

At the same time, other areas of structural biology, such as protein–ligand binding site prediction or the study of intrinsically disordered proteins, have seen less improvement. Notably, the field of drug design and its subfields – protein–ligand docking and ligand binding site (LBS)

prediction – is still largely confined to the use of rigid structures with the exception of a few movable side chains in the binding site. Similarly, the testing and training of LBS prediction software is often performed on a single holo structure as a representative, not taking into account the conformational diversity of the binding site or the rest of the protein. In the field of drug target identification, there are ongoing efforts to expand the druggable proteome by revealing binding sites that are normally obscured by conformational changes that occur between the apo and holo states.^{10,11} The PDB is our most valuable structural data resource, containing over 215,000 experimentally determined protein structures.¹ Many of these structures belong to the same protein but are solved under different conditions or bind different non-protein molecules; consequently, they often adopt different local or global geometries regardless of their identical sequence. This conformational variability, which can provide insight into sequence-independent folding dynamics, is not readily available to researchers who must extract, filter, group and analyze the structures of interest.

The use of apo and holo structures has been employed abundantly in areas that involve protein structure analysis. Najmanovich *et al.* compiled an apo-holo dataset with 221 proteins to observe the side-chain flexibility of different residues upon ligand binding.¹² LigASite was created to fill the gap of high quality apo-holo datasets by pairing biologically relevant interactions with their apo counterparts. It has been used in LBS detection studies and, to the authors' credit, remains available and features apo-holo pairs for 816 proteins since its last update.¹³ ComSin assembled 24,910 pairs of bound and unbound protein structures, clustered in three groups based on the sequence identity between the two forms (90, 95 and 100%), and added information about disorder and secondary structure to enable studies relating to disorder identification and characterization.¹⁴ PSCDB features 890 pairs of bound and unbound protein chains that were used to analyze the residue motions between the two states and classify them into 7 categories of motion types.¹⁵ CoDNAS features an extensive collection of protein structure clusters – referred to as conformers by the authors – collected by clustering PDB protein chains by 95% of sequence identity, but excluding homology from each cluster.¹⁶

Clark *et al.* compiled a dataset of apo-holo pairs featuring a minimum of two apo and holo representative chains for each of 304 proteins, to probe the flexibility of the two states by comparing them against their respective baselines, and revealed discernible differences in both RMSD and χ_1 side chain angles between the two states.¹⁷ They also used the dataset to assess the performance of five LBS predictors and found inconsistencies in the predictors' performance

among different structures of the same protein family that could not be attributed to structure quality criteria (i.e., resolution, Diffraction Precision Index,¹⁸ unobserved residues).¹⁹ Saldaño *et al.* compared AlphaFold models of 91 proteins with a curated set of apo-holo structure pairs and concluded that the models resemble the holo form 70% of the time, while proteins with larger documented conformational variability in the PDB produced lower quality AlphaFold models.²⁰ D3PM is a more recent database that collects and groups X-ray structures for 7649 proteins from the PDB into clusters based on their UniProt AC, annotates them according to their ligand-binding or non-binding status, and categorizes them into 4 types of overall structural changes and 5 types of pocket residue motions.²¹ Other notable works include AH-DB,²² CCProf,²³ PCDB²⁴ and the BUDDY-system,²⁵ all of which however, have ceased to exist.

Many of the existing datasets are manually assembled, a feature which indirectly limits their size and expandability. Current resources on the topic are by no means exhaustive and the need for large annotated datasets remains. Major leaps in machine learning have often been bottlenecked not by algorithmic logic, but by the size and availability of curated datasets.²⁶

Our initial motivation to assemble a dataset of apo-holo structures was indeed driven by the lack of sizable benchmarks on which we could test and improve our machine learning LBS predictor.^{27,28} The idea behind this is that testing and training on holo targets, while straightforward, is not a realistic scenario, as the holo pocket is an easy target for prediction – or docking – but often differs from the apo form of the same protein sequence.^{10,19}

We previously developed AHOJ to allow searching through the PDB for apo and holo targets, by providing a specific protein–ligand interaction, which in turn defines the protein pocket of interest.²⁹ In building the framework, it became clear that assigning apo or holo status to an entire chain, let alone a structure, can be vague and potentially misleading, in the presence of multiple binding sites on the same chain. In this sense, an apo or holo classification should be assigned based on a reference point that must be established first: a pocket or group of residues in a protein structure. A pocket-centric view is therefore practical for clarifying the apo-holo status.

While AHOJ's ad hoc search is useful for quick, parameterizable searches for any ligand in the PDB, processing thousands of structures to build a dataset remains computationally tedious. Furthermore, the search includes a number of options that users can customize, making it difficult to establish a baseline of standard results. Finally, since both AHOJ and its dependencies (PDB,¹ SIFTS,³⁰ UniProt²) are casually updated, it is beneficial to have a consistent and documented database.

Here, we describe a database built by matching the binding sites of biologically relevant protein–ligand interactions from the PDB with their apo and holo counterparts, considering PDB structures within the same UniProt AC (accession number). The biologically relevant interactions used as search input are taken from the redundant set of the BioLiP2 database,³¹ excluding peptide and nucleic acid ligands. In addition to the apo/holo classification that each binding site – or pocket – receives, several metrics are collected and computed for the query and mapped candidate pockets, including RMSD, solvent accessible surface area, molecular surface, allosteric factors, volume, b-factor, number of disordered residues and more.

Methods

Initial queries

The search and annotation of apo and holo states is started by a known protein–ligand interaction, which is itself a holo state by definition. We collected the biologically relevant protein–ligand interactions from BioLiP2 (redundant set, retrieved on 23.12.2023) and we discarded interactions between proteins and peptides or nucleic acids, and also interactions where the protein structure is not assigned a UniProt AC, since our methodology searches for candidates within a UniProt AC.

Search logic

We then conducted separate searches for each of these interactions, by specifying the ligand in the query structure, detecting its binding residues (i.e., pocket) based on a cut-off distance threshold (4.5 Å), and then mapping these residues – through SIFTS³⁰ – across other structures (candidates) that are assigned the same UniProt AC in the PDB, using an updated version of AHOJ.²⁹ The mapped residues are grouped into a mapped candidate pocket and used to detect ligands within the same cut-off distance threshold used to determine binding residues for the ligand of the query structure.

Search space

While query structures are not discarded from AHOJ-DB based on qualitative criteria, thresholds are imposed on the PDB structures that will be considered as possible candidates. However, the cut-off thresholds applied were restricted to allow a broad search that can be then refined through soft filters. The resolution cut-off threshold imposed on candidate structures was 7 Å, and the minimum percentage of query pocket residues mapped on the SEQRES of the candidate structure was set at 30%; otherwise, these structures were not processed at all. The

remaining structures were processed and their results were saved (global_results).

A soft filtration was applied upon all processed candidates, where the resolution threshold was 4.5 Å, and the results were sorted based on the following criteria mentioned in order of importance: percentage of mapped binding residues (MBR) on the candidate SEQRES, number of observed residues, resolution. These filtered results are saved in separate files and are the ones displayed in AHOJ-DB.

Apo/holo classification

The mapped candidate pockets are classified as apo or holo, depending on whether any ligands were detected in their vicinity, excluding solvent. It is worth noting that while we used biologically relevant interactions as starting points, we consider any detected ligand to be valid. The idea behind this is that even if these ligands are part of crystal packing contacts or crystallographic agents, they will affect the local geometry despite their questionable biological relevance. Modifications and filtering on these detected ligands may only be done by parsing the results and discarding, for example, ligands that appear in lists of crystallographic artifacts.

Metrics collected

Having access to the mapped candidate pockets which correspond to the same sequence part as the query pocket, enables comparisons beyond the apo/holo classification. For this reason, we collect an extensive number of metrics that can help discern observed differences in geometry between two states. We report these metrics in three groups: i) general metrics that concern the whole structure that the given pocket belongs to (i.e., resolution, experimental method used to determine the structure), ii) metrics that pertain to the protein structure chains that are involved in the formation of the mapped pocket – referred to as relevant chains – (i.e., TM-score³²), RMSD, allosteric factors like post-translationally modified (PTM) residues and ligands that occur within these chains) and iii) metrics directly related to the pocket (i.e., number of residues successfully mapped on the SEQRES, number of residues that are observed in the structure, solvent accessible surface area, molecular surface, volume,³³ b-factor of pocket atoms, radius of gyration of the pocket, distance between query pocket upon alignment, pocket alignment RMSD, orthosteric factors like other ligands, solvent molecules or PTMs that occur within the mapped pocket residues). More detailed information is included in the [Supplementary Information \(S.I.\)](#).

Assembling the database

The database consists of the results of all conducted searches. Each search is represented by an entry, within which, the results are listed in the form of mapped candidate pockets and their respective metrics and apo/holo classification. It is worth noting, that in line with the pocket-centric approach used, a candidate structure, or chain, may appear more than once in the results, if it has more than one pocket mapped onto its structure.

Web Server

The web server provides visualization and search capabilities and makes it possible to download the results through the graphical interface, or, provide the download links for a group of entries acquired through the search function in a CSV file.

Results

We processed 468,293 biologically relevant protein–ligand interactions from BioLiP2 without imposing any thresholds on the query structures, therefore considering all types of experimental structures as query structures. Candidates were subjected to filtering as described in the Methods section. Each biologically relevant protein–ligand interaction constitutes a separate entry and is referred to as such hereafter. It is worth clarifying that while the entry itself is a holo interaction, it is not considered in the holo results of the given entry. The entries can be retrieved by searching through AHOJ-DB for a PDB code, ligand Chemical Component Dictionary (CCD) name, UniProt AC, or a combination of these identifiers.

Custom sets

The filtering criteria specified in Methods were deliberately relaxed to allow for post-filtering and customization according to different use cases. Here we perform two different filtering operations and acquire two subsets of the database to illustrate this point. The resolution filtering also applies to queries.

- Subset 1
- Resolution 4.5 Å or better, only diffraction methods. MBR (SEQRES) filtering at 100%, no threshold for number of observed residues.
- Subset 2
- Resolution 4.5 Å or better, only diffraction methods. MBR (SEQRES) filtering at 100%, observed residues at 100% of MBR (zero tolerance for unobserved pocket residues).

To perform comparisons, we compute the average of each metric across the apo and holo results of each entry, and then perform a final aggregation to get the apo and holo grand

averages for all entries in the given subset. From these two grand averages for each metric, we compute the percentage difference between them by calculating their absolute difference and dividing it by the average of the two values. The comparisons below include entries with at least one apo and one holo result (besides the query).

Subset 1 yielded 451,408 entries across 27,162 UniProt ACs, out of which, 41% had at least one apo state as a result (185,193 paired entries). In UniProt ACs this translated to 39% having at least one apo state in the PDB (10,595 UniProt ACs). In this subset, while the entire query pocket is mapped onto the SEQRES of the candidates, it is not guaranteed that these residues are also observed in the structure; it therefore allows for comparisons in the number of unobserved residues between apo and holo structures. Indeed, in [Table 1](#), we see a 138% difference in the average number of unobserved residues (Unobserved MBR %) between apo and holo sites. This difference is further pronounced when considering the least disordered apo and holo pockets among an entry (see `unobserved_rsd_min` in [Table S1](#)), where the difference rises to 182%, indicating that apo pockets are less likely to have an ordered representation in the PDB compared to holo.

Subset 2 resulted in the same number of entries as Subset 1, since the filtering for observed residues is applied upon the mapped candidate pockets, and the number of entries with at least one apo result under these criteria was 175,388 (39%). The UniProt representation was 38% (10,262 UniProt ACs out of 27,162 had at least one apo candidate within structures of the same UniProt). Here the entire query pocket is mapped and observed fully in both apo and holo candidates which enables more confident pocket-related observations. A difference of 40% is observed in pocket RMSD, showing that apo pockets have consistently higher RMSD compared to holo when aligned onto the query pocket, hinting towards the existence of cryptic sites.

While less pronounced, this trend extends to chain RMSD – calculated by aligning the apo and holo relevant chains onto the respective query chains – (30% difference). There is also a noticeable difference in measured pocket volume with apo pockets appearing smaller overall (24% difference), which is less pronounced in total SASA values. It is worth noting that differences in SASA measurements seem to cancel out (see `sasa_min`, `sasa_max` in [Table S1](#)), indicating that averaging over all apo and all holo pocket values is not ideal for discerning differences related to volume or geometry through solvent accessibility; individual filtering per entry based on pairwise comparisons (apo pocket to holo pocket) could expose such differences more successfully.

Table 1 Selected metrics calculated for AHOJ-DB subsets 1 and 2 (both featuring 451,408 total entries – 96% of the total in AHOJ-DB). The number of entries with at least one apo and one holo result are 173,275 and 163,894 for Subset 1 and Subset 2 respectively, which were used to compute the aggregated metrics. Unobserved MBR % represents the percentage of mapped binding residues in the candidate SEQRES – MBR (SEQRES), that are not observed in the candidate structure.

Metric	Unit	Subset 1			Subset 2		
		Apo Average	Holo Average	% Difference	Apo Average	Holo Average	% Difference
MBR (SEQRES)	num.	12.16	12.16	0	11.91	11.91	0
Observed MBR	num.	11.86	12.1	2	11.91	11.91	0
Unobserved MBR %	%	2.44	0.44	138.41	0	0	-
b-factor per-atom	Å ²	51.29	39.07	27.06	50.6	38.66	26.74
SASA	Å ²	442.86	461.4	4.1	434.77	454.22	4.37
Volume	Å ³	114.32	153.28	29.11	117.53	148.95	23.58
Pocket RMSD	Å	1.47	0.96	41.51	1.43	0.95	40.13
Pockets	num.	45.26	131.88	97.8	44.68	132.16	98.93
Chain RMSD	Å	1.14	0.84	30.41	1.13	0.83	30.39
PDB chains	num.	44.63	129.33	97.38	44.08	129.89	98.65
PDB codes	num.	24.57	74.62	100.9	24.41	76.39	103.14

Certain trends persist throughout both subsets. The average number of pockets, structures, and chains per entry is consistently higher for holo states – 95% to 100% difference – throughout all subsets compared to apo, indicating an inherent bias within the PDB towards holo structures. This suggestion would be nicely complemented by an apo-centric search that is not present here; nonetheless, the metrics from the unobserved residues do point towards that direction. Per-atom average b-factor values (b-factor per-atom) are also in support of the case that apo pockets are less ordered overall (26% difference with holo pockets while resolution has an 8.6% difference – see resolution_avg in Table S1). For reference,

the number of entries that have at least one apo form in the entire unfiltered database is 230,269 out of 468,293 (49.2%), representing 13,068 out of 27,983 total UniProt ACs (46.7%).

From a ligand perspective

We grouped entries in Subset 2 by ligand and ranked them according to their abundance. The top 15 ligands are displayed in Table 2 (Total entries). The number of these entries that have at least one apo state in their results, are also shown (Apo-paired entries). While the overall average representation of apo states is about 30% of the total entries, the deviation between the

Table 2 Biologically relevant ligand interactions that are featured in Subset 2, grouped by ligand and sorted by number of interactions (sites). Two categories of sites are reported along with the respective number of unique structures, chains and UniProt ACs. “Total entries” refer to the total number of entries (holo interactions) featured in the subset, and “Apo-paired entries” refer to entries that include at least one apo state in their results. “Coverage” refers to the percentage of entries that were matched to at least one apo state out of the total entries.

Ligand	Total entries				Apo-paired entries				Coverage as %			
	UniProt	structures	chains	sites	UniProt	structures	chains	sites	UniProt	structures	chains	sites
ZN	4237	15,715	32,014	42,137	913	5355	9022	10,913	22	34	28	26
CA	3474	11,499	22,685	37,070	1220	5826	10,029	14,206	35	51	44	38
CLA	625	267	5147	29,637	14	35	88	252	2	13	2	1
MG	3852	10,638	24,237	26,922	1760	6185	13,121	14,657	46	58	54	54
HEM	984	5402	12,687	12,270	37	266	1293	882	4	5	10	7
MN	1097	3455	7554	10,614	446	1553	2972	4149	41	45	39	39
ADP	1287	3014	7955	7872	545	1496	4090	3869	42	50	51	49
FE	636	1948	5091	6129	126	585	1650	2165	20	30	32	35
CU	406	1685	3651	6003	117	693	1432	1799	29	41	39	30
SF4	591	1513	4332	5924	8	18	30	24	1	1	1	0
FAD	824	2648	5980	5818	28	91	225	202	3	3	4	3
ATP	916	2036	5911	5677	427	970	2500	2239	47	48	42	39
GLC	608	1610	2696	5521	380	1211	1926	4007	62	75	71	73
NAD	823	1686	5168	5072	215	561	1646	1670	26	33	32	33
GDP	548	1646	5671	4535	122	631	1498	1321	22	38	26	29

representation of different ligands is very large, ranging from 73% for glucose to 1% or less for chlorophyll A and iron sulfur clusters ($\text{Fe}_4 \text{S}_4$). This might indicate that certain ligands display an affinity for less ordered pockets or structures that are harder to resolve through diffraction methods in the absence of bound ligands.

While the majority of protein–ligand interactions are confined to a single protein chain, binding pockets also occur at the interface of chains. Such pockets are often neglected or misrepresented in datasets due to their complexity, but appear to warrant special attention.^{34,35} Particularly in apo-holo datasets, these pockets are even more difficult

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Search AHoJ-DB
Specify one or more of the following: [example1](#) [example2](#) [example3](#)

PDB_ID/s: UniProt_ID/s: Ligand/s:

Filters
☐ X-ray structures only
☐ Exclude NMR structures
 Resolution threshold: Å

[Search](#)

Search Results Summary

DB Entries	499
Unique PDB IDs	221
Unique UniProt IDs	1
Unique Target Ligands	193

[CSV](#)

Entry	Target PDB Chain(s)	UniProtID(s)	Resolution	Target Ligand	SASA	avg. SASA	Apo Sites	Holo Sites	
1a8i-A-GLS-998	1a8i / A	P00489	1.78	GLS (A_998)	256.54	17.1	10	247	View Download
1abb-A-IMP-920	1abb / AB	P00489	2.8	IMP (A_920)	752.13	53.72	120	134	View Download

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DB Entry: 1a8i A GLS 998

Query: 1a8i A GLS 998

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Found APO Sites / Structures: 10 / 10

Structure	AT	Res	MBR	RMSD	TM-sc	Lig	
7p7d / A	A	1.45	100.0	0.38	0.99		View
2gpn / A	A	1.99	100.0	0.18	0.99		View
3bd6 / A	A	2.0	100.0	0.58	0.99		View
1z6q / A	A	2.03	100.0	0.41	1.0		View
3bcr / A	A	2.14	100.0	0.56	0.99		View
2off / A	A	2.2	100.0	0.45	0.99		View
4mra / A	A	2.34	100.0	0.38	0.99		View
1z6p / A	A	2.4	100.0	0.35	1.0		View
2qn1 / A	A	2.401	100.0	0.57	0.99		View
2qn2 / A	A	2.698	100.0	0.55	0.99		View

Found HOLO Sites / Structures: 247 / 214

Structure	AT	Res	MBR	RMSD	TM-sc	Lig	
2qj4 / A	A	1.6	100.0	0.43	0.99	PLR	View
7onf / A	A	1.6	100.0	0.34	0.99	PLP VKK	View
5lrf / A	A	1.75	100.0	0.38	0.99	PLP KS3	View

Visualization:

BETA-D-GLUCOPYRANOSE SPIROHYDANTOIN
1A8i | Model 1 | Instance ASM_1 | B | A | GLS 998

Figure 1. Overview of AHoJ-DB interface – A) snapshot of the search page with the summary of results for a protein search. B) Example of detailed results and visualization of superpositions of selected apo and holo structures available under the “View” button.

to track and properly annotate across different structures. Allosteric effects are also closely related to structural changes, but existing resources are not readily available for machine learning applications.³⁶

Our approach is compatible with interchain binding sites and we therefore investigated how often the binding sites occur at the interface of two or more chains in our database. Depending on the distance cut-off threshold for determining binding residues, this percentage was 24% for multi-chain pockets and 9% for multi-UniProt pockets (heteromers), for a cut-off distance of 4.5 Å which is commonly used in cheminformatics.¹⁷ For comparison, in the redundant set of biologically relevant small molecule ligands in BioLiP2, over 10% of the interactions involve more than one PDB chain and about 6% include more than one UniProt chain. These differences can be explained by the stricter cut-off distance threshold used in the BioLiP2 database (van der Waals radii plus a tolerance of 0.5 Å).

Use cases

The database can be searched by three main modes: a) PDB code, b) UniProt AC, or c) ligand ID. The search by PDB code can be useful for gaining information about the diversity of known bound ligands, and the mode of binding. For example, searching for PDB code 1FNP of photosystem II from *Cereibacter sphaeroides* reveals 10 unique binding sites that bind 5 different ligands. It also clearly shows that 8 of these 10 binding sites appear at the interface of two proteins and interestingly, none of these binding sites is known in its apo form – one of the reasons could be that photosystem II is not stable enough to crystallize without a ligand. If we want to investigate this further, we can use the search by ligand name. One of the ligands identified in PDB: 1FNP is bacteriochlorophyll A (BCL) – data from our database shows that there are 3,005 unique BCL binding sites (as entries) from 189 structures, but there are just three apo structures for just one of these binding sites – it will be therefore difficult to study differences between apo and holo structures binding BCL.

The search by UniProt AC provides an overview of known binding sites of the protein and a number of biologically relevant ligands found. It can be compared to the information provided by the PDBE-KB³⁷; if we look at glycogen phosphorylase from *Oryctolagus cuniculus* (UniProt AC: P00489 – see Figure 1) we can see 499 unique protein–ligand interactions – these represent 193 unique ligands, while PDBE-KB provides information about 198 known ligands – five of those are not biologically relevant according to BioLiP2 database. The result also shows us that the most common ligand is PLP that is forming 284 unique interactions in 130 structures. There are 16 sites corresponding to PLP-binding sites in APO form –

it can be therefore used to study the variability of PLP binding and detect conformational changes between apo and holo structures. The results also show solvent accessible surface area calculated for the binding site (SASA), and the SASA normalized by the number of amino acids in the binding site that allows the analysis of the binding sites. More detailed information and visualizations of apo holo pairs are available through the “View” or “Download” buttons.

Conclusions

We have compiled what is to our knowledge the most comprehensive database for apo and holo structures to date. We adopt a pocket-centric perspective to avoid pitfalls in ambiguously assigning apo or holo status to entire chains or structures that have multiple binding sites. Here each entry represents a biologically relevant holo interaction, and therefore a protein binding site, which is considered the point of interest upon which the apo/holo classification is assigned to candidate sites. Furthermore, the database does not impose stringent constraints to allow post-processing customization of the data for different experiments.

The collected data show that apo structures are often not available in the PDB, which might hinder the construction of quality datasets for testing machine learning approaches. The data also suggest that binding sites at the interface of two or more chains account for about a quarter of known binding sites, and they should be considered and treated accordingly.

We believe that this database can support protein structure analysis as well as the development of machine learning based tools by integrating annotated structure ensembles that reflect the conformational variability present in the PDB. Some areas that stand to benefit from such a resource are drug design and docking experiments, testing and training of LBS predictors, validation of cryptic binding sites, detection of allosteric effects, investigation of order-to-disorder transitions and folding dynamics. We hope that this resource will help drive the new wave of drug design by allowing state-of-the-art algorithms to harness the power of curated, high-quality data.

CCrediT authorship contribution statement

Christos P. Feidakis: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization.
Radoslav Krivak: Validation, Software, Investigation, Formal analysis, Conceptualization.
David Hoksza: Writing – original draft, Supervision, Software, Resources, Data curation,

Conceptualization. **Marian Novotny**: Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Formal analysis, Conceptualization.

DATA AVAILABILITY

The data are freely available on the web server

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jmb.2024.168545>.

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ligands;
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drug design

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